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BOOK REVIEW

Hilbert space gone bananas (again)

Michael Janas, Michael E. Cuffaro and Michel Janssen: Understanding quantum raffles. Cham: Springer, 2022, 247 pp, €106.99 HB

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Understanding Quantum Raffles, according to its own preface, rests on three pillars: polytopes, philosophy and pedagogy. The book is inspired by Jeffrey Bub's *Bananaworld* (Bub, 2016). It was actually originally intended as a contribution to a special issue on *Bananaworld* and comes with a foreword by Bub himself (vii–xiv); however, the 'three Mikes,' as Bub calls them, have clearly crafted a stimulating read with a life of its own. The foundation on which the three pillars rest is a statistical concept of special relevance in quantum theory (QT): correlation.

Borrowing from Bub's works, the three Mikes start off explaining this by considering two chimps, 'Alice' and 'Bob,' who peel some rather curious bananas. These bananas can be oriented at three different angles relative to one another and either taste yummy or nasty. Taken individually, there is a 50–50 chance of each banana tasting either way. When Alice and Bob compare notes on bananas peeled, though, they find that one always tasted yummy and the other nasty if both bananas were peeled in the same orientation. This suggests that bananas are produced in pairs and that tastiness goes to only one member of the pair. However, Alice and Bob also find striking coincidences between *same* tastes when peeling along *different* directions. How much tastes should be *allowed* to so coincide can be quantified by a (here-violated) 'Bell-type inequality' (after Bell, 1964). Furthermore, bananas can be taken so far apart that no signal allowed by Einstein's relativity theories can be passed among them in time, and the findings remain the same. Therein lies the mystery: How on earth do these bananas arrange their tastes in these ways?

Of course, bananas are merely a representational aid for explicating a puzzle about (elementary) particles' *spins* and other properties. Raffles, the name-giving device of the book, are another representational aid for distinguishing different types of correlations. They correspond to baskets of tickets on which a range of possible 'settings' are specified, much like the peeling-orientations of the bananas. Raffles can also be 'mixed', so that one basket contains tickets with different possible

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settings (32). In this case, the two parties—Alice and Bob again—get to draw a ticket which is torn in half, and the two halves are distributed among them randomly. Finally, Alice and Bob each decide, at random and independently from one another, for a certain setting and receive an 'outcome' for it, which they note on their ticket. When they get together and compare their tickets, they note certain correlations between their outcomes and any correlation that can be simulated in this way may count as 'classical' (32).

Quantum correlations cannot be simulated in this way. Although this is well-known, the details are expounded on in the book. Furthermore, it is possible to design fictive devices, called 'Popescu–Rohrlich boxes,' that even exceed the quantum correlations (see Popescu and Rohrlich, 1994). The correlations actually exhibited by these different kinds of devices (particles, 'super-quantum' boxes, or raffles) are further illustrated, by the three Mikes, with the aid of *correlation arrays* that specify the probabilities of Alice getting one outcome given that Bob gets another, and which allow, for example, the formulation of a simple criterion for correlations being 'non-signaling' (26).

This gives a flavor of the book's pedagogy-pillar: the different representational aids allow the reader to see complicated things quite easily. Overall, I find the book to be an impressive achievement in this respect, though it is certainly too heavy for a stand-alone introduction. The most crucial device for illustrating the specialty of QT, however, is not the raffles but the first pillar: *polytopes*.

Many-faceted polytopes offer a way of characterizing correlations, such as those in a raffle, that goes back to Pitowsky (1989). These polytopes are generated by *anti-correlation coefficients* $\chi_{ab} \in [-1, 1]$ between settings a and b in a given raffle, or equally, in an actual quantum(-banana) experiment. The coefficients are introduced by the three Mikes as parameters in correlation arrays for variable setting choices (27), but their interpretation as quantifying anti-correlations is later justified by their connection to Pearson correlation coefficients (72).

For example, consider a setup with two spin-1/2 particles (say, electrons) that behave like the bananas discussed above. Using a variant of this setup due to David Mermin (1981), one can show (Sect. 2.5) that the possible anti-correlations exhibited by any raffle aimed at simulating the setup must lie within a tetrahedron, defined by sums over the anti-correlation coefficients χ_{ab}, χ_{ac} and χ_{bc} for settings (peeling directions) a, b and c, which sums must lie within the interval $[-1, 3]$ whenever either two or none of the coefficients are multiplied by -1.

This seemingly abstract and innocent criterion is really rather forceful: It allows for the quantum correlations to be characterized *in visible contrast* to classical and super-quantum correlations. This is because the correlations allowed by the quantum banana's peeling directions *exceed* the tetrahedron. They lie within an *eliptope* enclosing the tetrahedron, defined by the *eliptope inequality* (45):

$$1 - \chi_{ab}^2 - \chi_{ac}^2 - \chi_{bc}^2 + 2\chi_{ab}\chi_{ac}\chi_{bc} \geq 0$$

Remarkably, this inequality was discovered by Yule (1897) long before the advent of QT. Hence, while it is possible to derive the inequality 'from within' QT (Sect. 2.6), it is also possible to derive it 'from without' (Sects. 3.1–2). However,

the point is not that statistical correlations between certain variables *satisfy* this inequality: The remarkable thing is that quantum correlations *saturate* it, i.e., can reach its extremal value (0). Super-quantum correlations, furthermore, can reach points beyond the eliptope (see Fig. 2.8, 31).

Most intriguingly, it can be shown that the more different values of spin a system has available, the closer the given polyhedron for a potentially simulating raffle—which also becomes more complex and multi-faceted—approximates the full eliptope (see in particular Fig. 4.18, 155). Hence, it becomes harder and harder to exhibit the decisively quantum properties in systems with higher spins, and one may wonder what that tells us about the quantum–classical divide.

As can already be seen from the review so far, the book is a rich and impressive read. Both the pedagogy and polytope pillars are fairly massive, and I hope to have offered at least some orientation for the interested. Admittedly, I sometimes found it difficult not to get lost in the technicalities, and maybe the representational richness even adds to this.

Much like the book itself, I will now conclude by finally considering the second pillar: *philosophy*. There are many interesting suggestions as to what it all means, which are discussed in a penultimate chapter. Included are discussions of the distinction between principle-theoretic and constructive approaches to theorizing, 'big' and 'small' measurement problems, and a crucial distinction between dynamical and kinematic aspects of a given theory. Kinematic aspects are concluded to acquire a special relevance in QT: They encode "generic constraints on the possible values of observables as well as on the correlations between such values" (187). Observables, however, are not taken to represent the properties a given system has independently of the context of a measurement-interaction (see 185–6).

I was indeed most curious about the philosophical part of the book, but I felt that the resulting picture looked more like the outline of a—very interesting—research program than a fully developed conception. For instance, the three Mikes compare their approach to 'epistemic' (e.g., QBism or Copenhagen) and 'ontic' (e.g., Bohm, collapse, and Many-Worlds) approaches. In their own approach, "the status of the state vector is epistemic rather than ontic" (1; emphasis omitted), but at the same time, "the quantum state description… is a window into the underlying *non-Boolean* structure of *the world*" (218; first emphasis in the original, second mine). That sounds fairly 'ontic' to me, and I would have liked to understand the suggested ontology (if any) better.

Maybe this must all be understood in terms of the "broadly Kantian outlook" (xvii) shared by the three Mikes, and so 'the world' here is in no small part defined by the epistemic conditions of a conscious experiencer. As far as I can tell, however, the name 'Kant' only reappears explicitly in the references, and so the exact inspirations taken from Kant, as well as the exact mix of epistemology and ontology, remained a bit unclear. That is a pity; I would have enjoyed seeing these Kantian inclinations put to work in more detail (see Boge 2018, 2021).

Nevertheless, I hope to have shown the many stimulating and fruitful ideas contained in the book. Anyone interested in 'broadly epistemic' approaches to QT and in understanding quantum correlations, the quantum–classical divide, and connected topics should give it a read.

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